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Musculoskeletal Injury Risk After Sport-Related Concussion

A Systematic Review and Meta-analysis

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Background: Clinical management of sport-related concussion typically involves a symptom checklist, clinical examination of mental status, and neurocognitive testing. However, recent studies have identified unresolved, impaired sensorimotor function after athletes return to sport. A review and meta-analysis of all current literature regarding risk of subsequent musculoskeletal (MSK) injury after concussion has yet to be published in the medical literature.

Purpose/Hypothesis: To determine the odds that athletes will sustain MSK injury after concussion. It was hypothesized a priori that concussion would increase the risk for MSK injury.

Study Design: Systematic review and meta-analysis.

Methods: PubMed and Google Scholar were searched from January 2000 to November 2017. Reference lists of the included studies were manually searched. Two reviewers independently searched the literature for studies published in English that reported MSK injury after athletes returned to play following a concussion. Two independent reviewers completed data extraction using PRISMA guidelines and assessed study quality using the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies from the National Institutes of Health. Random effects meta-analyses were used to calculate odds ratio (OR) and incidence rate ratio (IRR) of MSK injury after concussion. The primary study outcome of interest was the number of athletes who sustained MSK injury after concussion.

Results: Eight studies met inclusion criteria for meta-analysis. Meta-analysis results indicated that athletes who had a concussion had 2 times greater odds of sustaining a MSK injury than athletes without concussion (OR, 2.11; 95% CI, 1.46-3.06). In addition, athletes with concussion demonstrated a higher incidence of MSK injury after return to sport compared with nonconcussed athletes (IRR, 1.67; 95% CI, 1.42-1.96). Further analysis showed that both male and female athletes with concussion were at an increased risk of MSK injury compared with their respective same-sex, nonconcussed controls (OR $>$ 1.56, $P < .01$).

Conclusion: Based on the evidence of higher risk of MSK injuries after concussion, standard clinical assessments for athletes with concussion should include not only physical symptoms and cognitive function before return to sport but also neuromuscular risk factors associated with increased risk for MSK injuries.

Keywords: sport-related concussion, musculoskeletal injury risk, lower extremity

The Centers for Disease Control National Center for Injury Prevention and Control reported that traumatic brain injury (TBI) accounted for 2.8 million emergency department visits in 2013 and affected individuals from newborns to people older than 75 years.[41] Nonfatal TBIs in individuals younger than 20 years secondary to sport and recreational activities account for 248,418 emergency department visits in 2009.[33] Many sport-related TBIs, however, are unrecognized and do not result in emergency department visits. Therefore, a realistic estimate of sport-related TBIs is somewhere between 1.6 and 3.8 million incidents per year.[22] Concussion, classified as a mild TBI, was defined by the panel of experts at the 4th International Conference on Concussion in Sport in Zurich as a complex pathophysiological process that affects the brain and is induced by biomechanical forces.[30] Given the high prevalence of the condition, the increased risk for a second concussion upon return to sport (RTS),[28] and potential long-term sequelae that range from increased risk of mild cognitive impairment to severe neurodegenerative disease,[27] a tremendous amount of effort and research have been allocated and completed to date on concussion. Thus, systematic reviews on topics relevant to sport-related concussion are warranted to gather sufficient evidence and distribute the pooled conclusions to medical professionals.

Clinical management of sport-related concussion usually involves several domains such as symptom checklist; clinical examination of mental status; behavior, sleep, eye, and muscle disturbance; postural stability assessment; and

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neurocognitive testing.[2,30] Some individuals with sport-related concussion exhibit neuromuscular characteristics similar to those of individuals with neck pain and whiplash in terms of the visual-vestibular symptoms, tension headache, balance disturbance, and proprioceptive deficits.[5,21,23,44] Independent of the actual mechanism of TBI, these previous findings indicate that sensorimotor function is disrupted in individuals with concussion. A primary role of the sensorimotor system is to convey accurate sensory information regarding visual, vestibular, and somatosensory afferent to the central nervous system to effect the motor functions necessary to achieve functional joint stability with feedback and feedforward controls.[39]

Impaired sensorimotor function may have deleterious effects on other aspects of athletic performance. Several recent studies have investigated the effect of concussion on musculoskeletal (MSK) injury.** To the authors' knowledge, the medical literature does not contain a systematic review and meta-analysis looking at MSK injury risk after a sport-related concussion. Thus, the primary aim of this systematic review and meta-analysis was to determine the odds and incidence rate ratios of MSK injury after sport-related concussion. Further, meta-analyses for athletes with concussion were conducted for males and females separately. It was hypothesized that concussion would increase the risk for MSK injury.

METHODS

Search Strategy

A review protocol was established before beginning the literature search. Two reviewers independently performed electronic searches in PubMed and Google Scholar to identify relevant articles. Search terms were mapped to relevant terms by use of the Medical Subject Headings (MeSH) database of articles from PubMed. Two concepts were entered into the database: Concept 1, "cognition," "brain," "brain injuries"; Concept 2, "athletic injuries," "lower extremity." Concepts were grouped using the "AND" operator. A manual search was performed on the reference lists of the relevant papers. Publication details from all identified studies in the literature search were exported to bibliographic software. Results of the literature search are shown in Figure 1.

**References 3, 4, 6, 8, 10, 12, 15, 24-26, 34-36.

Selection Criteria

The following criteria were used during the literature search: (1) studies published between January 2000 and November 2017, (2) studies published in English, and (3) studies that reported MSK injury risk of athletes after a diagnosis of sport-related concussion. Review articles were excluded. Titles and abstracts were first reviewed by selection criteria. Two reviewers independently applied the selection criteria, and consensus was reached to resolve any discrepancies.

Quality Assessment

Included studies were assessed for methodological quality using the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies from the National Institutes of Health, which is composed of 14 items.[37] All studies were assessed independently by 2 reviewers, and discrepancies were resolved by consensus. The purpose of the quality assessment was to identify characteristics of study quality and design for all included studies and to provide a qualitative description of the study characteristics. For all items in the assessment, 2 independent variables were considered: (1) concussion diagnosis and (2) athlete exposures upon RTS. For item 7, a follow-up time frame of 365 days was considered sufficient to see an association between concussion exposure and injury outcome. For item 9, the 2 independent variables addressed were concussion or control and athlete exposure (ie, time). To satisfy item 9 and be scored as a 1, both exposures had to be satisfied.

Data Extraction and Synthesis

A data-extraction form designed for this review was used to obtain data from each of the studies. The primary outcome of interest was the number of athletes who sustained MSK injury after RTS from a concussion. The number of athletes was used to calculate risk to retain homogeneity of the extracted data. With the exception of the sex breakdown analysis, if an athlete sustained multiple injuries during the follow-up time period, the athlete was considered 1 injured athlete in the current analysis. For the sex-specific analysis, 1 study reported the number of MSK injury as cases, where 1 athlete could contribute more than 1 injury to the data.[15] Athlete exposure was noted when provided in a study. Authors were contacted for additional data or clarification of data presented in the manuscript.[3,8,15,24]

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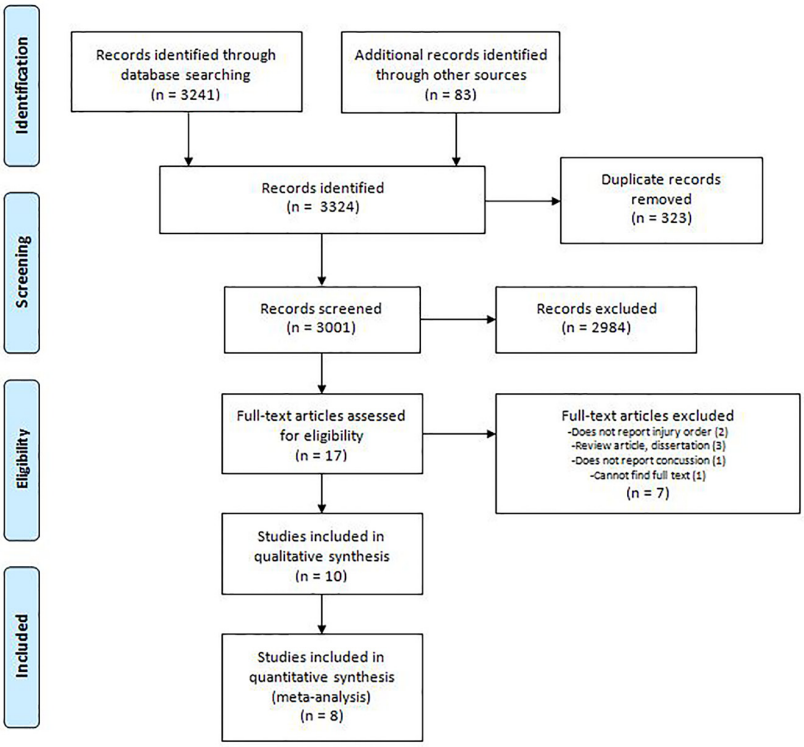

Figure 1. Article selection, following PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2009 flow diagram.

Statistical Analysis

Data were analyzed by use of StatsDirect software (v 2.8) and RevMan (v 5.3; The Nordic Cochrane Center, The Cochrane Collaboration, 2014). The numbers of athletes injured after concussion were combined via odds ratio (OR) and incidence rate ratio (IRR) meta-analysis through use of a random effects model. Odds ratio represents probability, where an OR value greater than 1 indicates that the odds of MSK injury is higher in athletes who have sustained a concussion compared with those who have not.[43] Incidence rate ratio is the incidence rate of MSK injury, the outcome measure, in the concussed group divided by the incidence rate of MSK injury in the unexposed group.[43] Odds ratio meta-analyses were also performed for each sex to determine the odds of injury after concussion compared with control athletes. Differences in risk were considered significant if $P \leq .05$ and the 95% confidence interval (CI) did not contain 1. The I^2 index was used to assess the heterogeneity of the results of the included studies, where variability due to heterogeneity is represented by a larger I^2 index.[18]

RESULTS

A total of 3324 studies were identified in the databases and additional sources, from which 17 studies were identified for full-text assessment. Ten studies were mutually agreed upon for inclusion by 2 reviewers (Table 1). All studies were prospective or retrospective cohort methodological designs, indicating that they were level 3 evidence studies.

Quality Assessment

The number of items satisfied in the quality assessment checklist ranged from 6 to 11 (of a possible 14). The quality assessment revealed that 5 studies inadequately reported or measured athlete exposures[3,4,8,25,34] and 5 studies used a short postconcussion timeframe to see an association between concussion and injury.[3,15,25,26,35] Although healthy, noninjured controls matched for potential confounding factors (ie, sex, age, position) were used in 5 studies,[3,8,15,24,26] several reported that exposure levels were not the same between the concussed athlete group and matched controls. Three studies used a control group composed of athletes who sustained a nonconcussive MSK injury, and the athletes were tracked for new injuries after RTS.[4,34,35] The study of high school athletes did not compare injury risk against a control group; rather, the study developed generalized estimating equations to test whether concussions predicted subsequent injury.[25] Most authors cautioned against the generalizability of their results to other populations of athletes of different sport, age, and playing level and noted small sample sizes.[3,6,15,24,26,34,35] Sample sizes of the concussed athlete group varied between studies used in the quantitative synthesis, ranging from 44 athletes[24] to 2004 athletes.[25]

TABLE 1
Study Characteristics[a]

Lead Author (Year)	Athletes With Concussion, No. and Level of Participation	Sports	Control Athletes	Quality Checklist Score	Time From Concussion to Return to Sport[b]	Injuries Tracked	Injury Tracking Time Period
Brooks (2016)[3]	75 (58 M, 17 F) College	Football, soccer, hockey, softball, basketball, wrestling, volleyball	Nonconcussed, matched players (sport team/sex, exposure during 90-day return to play, position)	8	21 days	Acute, noncontact, lower extremity, musculoskeletal injuries	90 days
Burman (2016)[4]	281 (206 M, 75 F)	Ice hockey, soccer, floorball, handball	Athletes with an ankle sprain, same 4 sports	7	Not reported	Not defined	24 months
Cross (2016)[6]	135 (135 M, 0 F) Professional	Rugby	Nonconcussed players	9	11 days	Any injury	2 seasons
Fino (2017)[8]	110 (76 M, 34 F) College	Baseball, basketball, lacrosse, football, soccer, softball, swimming, tennis, wrestling, volleyball	Nonconcussed, matched players (sport, position, exposure/skill level, history of lower extremity injuries 1 year before their matched concussion, age, height, weight)	9	18 (±40) days	Acute lower extremity injuries	12 months
Herman (2017)[15]	73 (52 M, 21 F) College	Football, basketball, soccer, lacrosse	Nonconcussed, matched players (sport, starting status, main position)	9	9 (±7) days	Acute, lower extremity, musculoskeletal injuries	90 days
Lynall (2015)[24]	44 (28 M, 16 F) College	Cross country, field hockey, football, lacrosse, soccer, swimming, rowing, softball, wrestling, basketball, tennis	Nonconcussed, matched athletes (sex, sport, competition playing time, age, height, weight)	11	Not reported	Acute, lower extremity, musculoskeletal injuries	12 months
Lynall (2017)[25]	2004 High school	Baseball, basketball, crew, cross country, field hockey, football, golf, gymnastics, track, lacrosse, soccer, softball, swimming and diving, tennis, volleyball, wrestling	Not applicable	6	Not reported	Acute and chronic, lower extremity, musculoskeletal injuries	Undefined
Makdissi (2009)[26]	117 (117 M, 0 F) Professional	Australian football	Nonconcussed, matched players from same team (position, age, size)	10	Not reported	Any injury	Undefined
Nordström (2014)[34]	66 (66 M, 0 F) Professional	Soccer	Nonconcussed injured player	9	Not reported	Any injury	12 months
Nyberg (2015)[35]	81 (81 M, 0 F) Professional	Hockey	Athletes diagnosed with medial collateral ligament tear or knee distortion	9	Not reported	New traumatic injuries	42 days

[a]F, female; M, male.
[b]Data are reported as mean (±SD).

Demographic Data

A total of 3027 athletes with concussion and 4226 control athletes were studied in the 10 articles included in the review. One study[25] did not report sex breakdown of the subject population. The remaining 9 studies in the qualitative synthesis did provide sex breakdown of the subject population, reporting 860 male and 163 female athletes with concussion compared with 3719 male and 507 female control athletes without concussion. Athletes with concussion ranged from 19.8 (SD, ±1.3)[3] to 25.3 (SD, ±4.2)[34] years of age, with a similar range for the control athletes. Sport and participation levels for each study are included in Table 1. Eight studies reported distribution of sports for the subject population.[4,6,8,15,24,26,34,35] In the group with concussion, 26% participated in soccer, 15% ice hockey, 15% Australian Rules Football, 13% rugby, and 11% American football, and the remaining 20% (each sport <5%) of the athletes participated in floorball, handball, baseball, basketball, lacrosse, softball, swimming, tennis, wrestling, volleyball, cross country, field hockey, or rowing. For the control group without concussion, the distribution of sport participation was different; 57% participated in soccer, 16% rugby, 9% floorball, and the remaining 18%

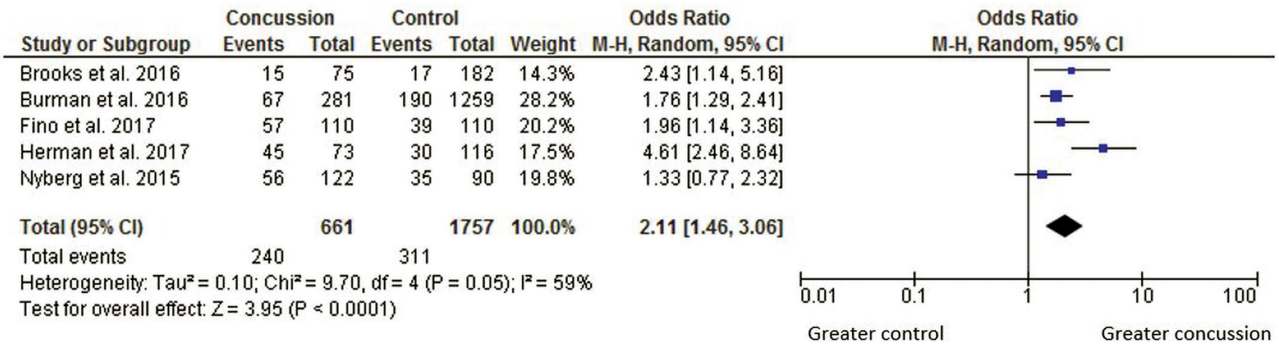

Figure 2. Forest plot of odds ratio of injury after concussion. The mean odds ratio and 95% CI data from individual studies in addition to the pooled data are shown. Odds ratio (OR) > 1 indicates greater odds of injury for athletes with concussion compared with control athletes.

(each sport <5%) ice hockey, handball, baseball, basketball, lacrosse, American football, softball, swimming, tennis, wrestling, volleyball, cross country, field hockey, rowing, or Australian Rules Football.

Individual Study Results

In a study of college athletes, Brooks et al[3] reported an increased incidence of MSK injury in athletes with concussion in the 90-day RTS period (OR, 2.48; 95% CI, 1.04-5.91). Another study of college athletes reported an increased instantaneous risk of lower extremity MSK injury in the year after RTS (hazard ratio, 1.67; 95% CI, 1.11-2.53).[8] Lynall et al[24] reported a higher rate of lower extremity MSK injuries in athletes with concussion at both 180 days and 1 year after RTS (IRR, 2.02; 95% CI, 1.08-3.78; IRR, 1.97; 95% CI, 1.19-3.28). Herman et al[15] similarly reported 3.39 times higher odds of lower extremity MSK injury after concussion (95% CI, 1.90-6.05). Whereas the 4 aforementioned studies evaluated injury risk in a collegiate athlete population, only 1 study investigated the risk of lower extremity MSK injury after concussion in high school athletes. For every previous concussion, the authors reported a 34% higher odds of sustaining a time-loss injury (OR, 1.34; 95% CI, 1.13-1.60).[25] Interestingly, the researchers observed an increased risk only for time-loss injuries, which was defined as an injury that resulted in restriction from athletic participation beyond the day of injury, but not for non-time-loss injuries or for all injuries combined. Professional rugby players were reported to have a 60% greater risk of injury after concussion (IRR, 1.6; 95% CI, 1.4-1.8).[6] In professional European soccer players, concussion was associated with increased risk of injury in the year after RTS (hazard ratio, 4.07; 95% CI, 2.14-7.76).[34] A study of Australian Rules Football players did not find a significant difference in injury rate between athletes with concussion (7.25; 95% CI, 3.48-13.33) and control athletes (3.25; 95% CI, 1.96-5.07).[26] Similarly, elite ice hockey athletes with concussion did not have an increased risk for injury compared with a control group of athletes who returned to sport from a knee injury.[35] In a study of athletes of unspecified participation levels, athletes with concussion were more likely to sustain injuries than the control group of athletes who had returned to sport from an ankle injury in a 2-year follow-up period (OR, 1.72; 95% CI, 1.26-2.37).[4]

Odds Ratio: Injury Risk After Concussion

Athletes with concussion had 2.11 times greater odds of sustaining a MSK injury compared with control groups (OR, 2.11; 95% CI, 1.46-3.06; $P < .001$; $I^2 = 59\%$) (Figure 2). The meta-analysis for the OR included 661 athletes with concussion and 1757 control athletes.

Incidence Rate Ratio: Rate After Concussion

Athletes with concussion in this analysis experienced 197 events in 1471 athlete exposures, whereas athletes in the control group had 15,821 events in 37,488 athlete exposures. These incidences per exposure were included in the meta-analysis for the incidence rate ratio. Athletes with concussion were 1.67 times more likely to experience a lower extremity MSK injury than control athletes (IRR, 1.67; 95% CI, 1.42-1.96; $P < .0001$; $I^2 = 0\%$) (Figure 3).

Odds Ratio, Stratified by Sex

Odds ratio meta-analysis of injury risk stratified by sex was performed using the 4 studies[3,4,8,15] that reported the number of athletes by sex in both the concussed and control groups. Male and female athletes with concussion both demonstrated an increased likelihood of MSK injury compared with control athletes of the respective sex. Female athletes with concussion had 2 times greater odds to sustain injury compared with nonconcussed female control athletes (OR, 2.23; 95% CI, 1.40-3.57; $P = .001$; $I^2 = 0\%$). Male athletes with concussion similarly demonstrated an increased odds for MSK injury compared with nonconcussed male control athletes (OR, 2.06; 95% CI, 1.26-3.37; $P = .004$; $I^2 = 62.9\%$). We found no significant difference in MSK injury risk between male and female athletes with concussion (OR, 1.53; 95% CI, 0.86-2.70; $P = .15$; $I^2 = 34.6\%$).

Study or Subgroup	Concussion Events	Concussion Exposures	Control Events	Control Exposures	Weight	Incidence Rate Ratio D-L, Random, 95% CI
Brooks et al. 2016	15	448	17	1084	5.5%	2.13 [0.99, 4.54]
Cross et al. 2016	119	976	1398	18500	75.4%	1.61 [1.33, 1.95]
Makdissi et al. 2009	10	138	19	585	4.5%	2.23 [0.93, 5.04]
Lynall et al. 2015	50	14259	37	17319	14.6%	1.64 [0.05, 2.58]
Total (95% CI)		**1471**		**37488**	**100.0%**	**1.67 [1.42, 1.96]**
Total Events	194		15821			

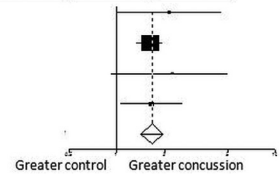

Figure 3. Forest plot of incidence rate ratio after concussion. The mean incidence rate ratio and 95% CI data from individual studies in addition to the pooled data are shown. Incidence rate ratio (IRR) > 1 indicates greater incidence of injury for athletes with concussion compared with control athletes.

DISCUSSION

The results of the current meta-analysis demonstrated that athletes who sustained a sport-related concussion had a significantly higher odds of MSK injury than nonconcussed control athletes. Furthermore, male and female athletes both demonstrated an increased risk for subsequent MSK injury after a sport-related concussion compared with control athletes of the same sex.

The current review and meta-analysis clearly demonstrated an increased risk of MSK injury after sport-related concussion. Two studies included in the current systematic review did not use matched, nonconcussed healthy athletes as a control group; rather, those studies used athletes with a reference injury and tracked the RTS time period after the injury (ankle sprain, knee injury).[4,34] This presents an interesting point of comparison to nonconcussed control athletes, since level 1 evidence from a published systematic review supports that previous MSK injuries increase the risk of reinjury to the same limb or new injuries to the contralateral side.[9] A higher risk of injury was observed in the concussed group compared with the control group before the concussion event.[4] Conversely, 2 studies identified no increased injury risk or rate between the concussed group and control group before the concussion event.[6,24] Two studies accounted for the previous lower extremity injury in the data analysis and reported greater risk of injury after concussion when adjusted for injury history.[8,34] If the history of concussion is a greater risk factor for future MSK injuries, future research must investigate the concussion-induced changes and ensure that athletes are prepared for RTS.

Two studies were excluded from the systematic review because of the methodological reporting of the data.[10,36] These studies were surveys that inquired the number of concussions as well as number of injuries players experienced during their athletic career. Although both studies reported association between concussion and injury, they did not report the order of concussion and lower extremity injuries. Because these 2 studies did not report the sequential order of concussion and injury, the risk of injury after concussion could not be specifically determined. However, both studies demonstrated an association between the number and frequency of concussions and the number and frequency of lower extremity injury, which further supports the finding of this meta-analysis.

Physiological Reasoning Behind Increased Injury Rate After Concussion

Concussion involves several clinical domains: symptoms, physical signs, behavioral changes, cognitive impairment, and sleep disturbance.[29] Physical signs are one domain commonly affected by concussion that can resolve more quickly, while other concussion domains have persistent impairments. One study reported that intracortical inhibition within the primary motor cortex might be present more than a year after physical concussion symptoms have subsided.[42] Impaired neural activity has been suggested to affect several aspects of motor control, leading to decreased neuromechanical responsiveness.[32] For example, decreased maximum strength has been reported after concussion.[38] Concussion can also affect the cervical spine proprioception, vestibular-ocular system, and visual systems that result in decreased postural stability.[5] Poor postural stability has been associated with ankle sprain, one of the most common lower extremity MSK injuries in sports.[19] Persistent sensorimotor impairments after resolution of concussion symptoms would likely contribute to an increased injury risk, and continued studies are warranted.

Clinical Implications

Clinicians should be aware of the greater risk for MSK injury after concussion. To minimize sequelae from the concussion, clinicians should consider the current evidence to determine RTS after a concussion and use previously established guidelines and protocols. For the studies included in the meta-analysis, RTS after concussion ranged from an average of 9 (SD, ± 7)[15] to 18 (SD, ± 40) days.[8] Persistent neurophysiological changes from a concussion, such as decreased electrophysiological responses and intracortical inhibition, may lead to impaired neuromuscular function.[13,42] Cognitive impairments are examined with a computerized neurocognitive test at baseline and after concussion and often contribute to the RTS timeline.[31] After a concussion, deficits in reaction time and processing speed

were demonstrated with neurocognitive testing.[20] These impaired cognitive domains may affect an athlete's safety and performance, as delayed neural processing of visual input or reduced field of vision could affect the athlete's ability to anticipate changes in external conditions and loads and to adapt movement accordingly. Using a validated computerized cognitive test, a prospective study found that individuals who subsequently sustained noncontact anterior cruciate ligament (ACL) injury had significantly slower reaction time and processing speed and reduced visual and verbal memory compared with individuals who did not experience any MSK injuries.[40] A study of NCAA (National Collegiate Athletic Association) Division I football players identified, with 74% sensitivity and 51% specificity, that lower neurocognitive reaction time was an indicator of elevated risk for lower extremity strains and sprains.[45] Together, these 2 studies present preliminary prospective evidence of the relationship between cognitive function and injury risk.

While 5 studies did not explicitly define the MSK injuries tracked after RTS from concussion, 4 studies stated that acute lower extremity MSK injuries were recorded.[3,8,15,24] All 4 studies reported increased risk of injury in the concussed athlete group compared with the control group. In addition, Makdissi et al[26] reported that 8 of the 10 injury events in the follow-up period were lower extremity MSK injuries, and Lynall et al[25] included both acute and chronic lower extremity MSK injuries in their analysis that reported increased odds of time-loss injuries after concussion. These findings, in combination with the findings of the current review and meta-analysis, show that increased risk of lower extremity MSK injury can be associated with concussion history. Female athletes are known to be at higher risk of time-loss acute lower extremity injury, such as an ACL tear.[1] Since the current meta-analysis data show that female athletes with concussion are at an increased likelihood of MSK injury compared with female athletes without concussion, extra caution should be taken in RTS. The potential sustained effect of concussion on sensorimotor pathways could exacerbate faulty neuromuscular movement patterns that are known to increase female athletes' risk of injury.[17]

Alterations in kinematic and kinetic variables during movement have been extensively studied and identified as ACL injury risk factors.[16,17] A recent observational cohort study showed that lower extremity stiffness is altered after a concussion in NCAA Division I football players, which could contribute to injury risk.[7] In addition, it has been reported that drop-jump landing biomechanics, a task frequently used to assess risk for ACL injury, varies with baseline neurocognition.[14] Healthy athletes without concussion who had lower neurocognitive performance demonstrated movement patterns associated with increased risk of ACL injury.[14] Thus, based on the evidence presented of higher risk of MSK injuries after concussion, screening for neuromuscular risk factors associated with lower extremity injury should be incorporated into RTS protocols from concussion.

This review is not without limitations. Only studies in English were included in the search criteria, which could potentially introduce language bias. Furthermore, several studies were retrospective level 3 evidence studies. A greater number of level 1, prospective studies would increase the level of evidence presented in the current review. Sample sizes varied between studies; however, even the study with the smallest sample size (n = 44) detected a significant difference in injury rate between the athletes with concussion and matched control athletes without concussion a year from RTS.[24] A small sample size and small number of overall MSK injuries compared with the known number of concussions is a limitation of the meta-analysis. In addition to a larger sample size, information regarding the specific mechanism of injury could allow for investigation of more specific questions. Previous studies have identified that 70% of ACL injuries are due to a noncontact mechanism.[11] Therefore, future studies investigating the risk of lower extremity injury after a concussion should try to include this information, as this could help better identify the causes of increased risk of injury after a concussion.[3,8,34,35] Previous studies have reported a higher risk of acute-onset and more severe MSK injuries after a concussion. Future studies could help identify the complex relationship between concussion, MSK injury mechanism, and the severity of injury.

Last, because of the limited number of studies and available cases, it was not possible to evaluate the effects of sport types on the risk of MSK injuries in athletes following concussion. A study of ice hockey players did not find significant increases in MSK injuries after a concussion.[35] This finding is likely due to a higher proportion of MSK injuries from direct contact to other players and the wall. In addition, a study of Australian Rules Football players did not report a statistically significant difference in injury rate between athletes with concussion and those without, despite there being twice the likelihood of subsequent MSK injury in the concussed group (OR $>$ 2).[26] Future studies with sufficient sample sizes should be conducted to elucidate postconcussion risks of MSK injury between different sports. This information would help guide RTS protocols unique to each sport and risk factors associated with RTS.

CONCLUSION

In this meta-analysis, risk of MSK injury increased after a sport-related concussion. The evidence presented of higher risk of MSK injuries after sport-related concussion indicates that screening for neuromuscular risk factors associated with MSK injury should be incorporated into RTS protocols following concussion.

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